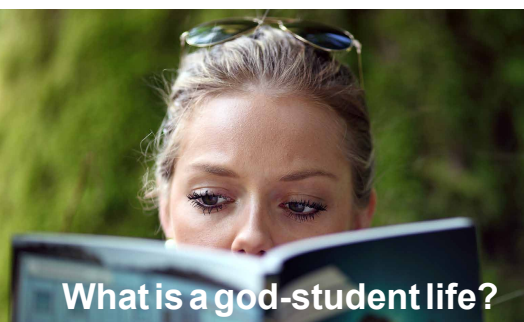


# Assisting First Year Students Maintain a Prosperous Student Life

It is important to understand “student life” and the difficulties that students face to help them develop good study habits. A summary of responsible groups that can assist students has been described below. These groups include lecturers for first-year students, group assistants and fellow-students. The importance of extra-curricular activities is also highlighted.

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Is this a good student life?

How to have a meaningful student life?



Maybe yes, but is it good enough?  
Is something missed?



## Institution's responsibilities

- Arrangements and orientation in the beginning of the semester
- Good communication with lecturers in order to follow up first-year students

## Lecturer responsibilities and roles

- To create happiness
- To create and develop curiosity
- To supervise or assist with supervision
- To care and support students and create a safe environment

## Student-assistant responsibilities

- Correcting and giving feedback on student obligatory assessments
- Following up students and encourage students to attend in groups and give them feed back during session.

## Fellow-students responsibilities

It is essential that students know that they are there to assist in learning. Just like a football team's success which depends on engagement and needs all players to be active and help each other.



## Students responsibilities

Students should be aware of their tasks and take responsibility for their own learning.

Students can contribute to their own learning and to the development of a campus climate in which all can grow and learn.

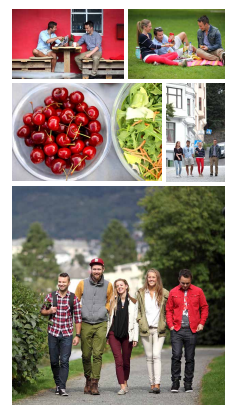
To keep students motivated, interested and active during first semester in mathematics

My experience during nine years lecturing for MNT students confirms that it is important to make students aware of the following in the beginning av semester

Experience in recent years also confirms that it was important to make students aware of:

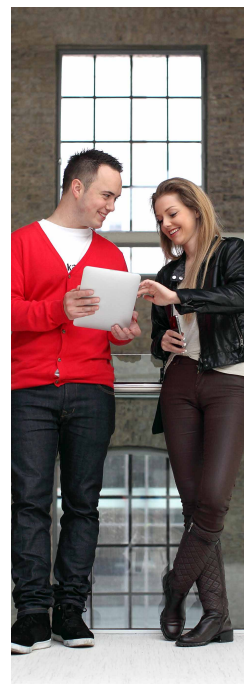
- study skills, their responsibility for their own learning and the importance of acquiring the ability to reflect on their own learning, already in the first lecture.
- the importance of responsibility for their own learning and emphasize the importance of reading systematically and actively participate in the planned teaching activities (the lectures, refresher classes, seminars and task bill hours).

My experience shows also that if students thrived in the lectures, and succeeded in the problem solving hours, promoted an interest and passion in learning which in turn helped to increase



their efforts and awareness of their own learning, their own study methods and reflection skills. The result was then ideal heightened sense of achievement and satisfaction, which had a positive effect on learning, which in turn helped to influence the academic performance.

Changes in courses structure included assessment, additional brief classes for weak candidates and increased number of compulsory assignments. Lecturer and group leaders also tried to adapt to students' needs and attempted to engage the less motivated students to work systematically through the entire semester.



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# Maths + Talk<sup>2</sup> = Methods on Promoting a Culture of Learning

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## Abstract

The purpose of this case study is to better understand the effect of team-based problem solving (TBPS) and the impact on changing the culture of learning and knowledge building in mathematics. When lecturers present new subject material, students then develop their understanding of the content by discussing and explaining the material to other students in small groups. This method was implemented for *Subsea and Marine Technology Engineering* students at Bergen University College over the course of several years (2009 - 2015). The learning from this project revealed that students develop a stronger understanding of mathematics.



It is essential that students know how to approach an effective learning process, and they can improve fellow students' learning and they are there to make that happen, like a football team's success depends on active and helpful team engagement by all players.



## Talking the Language of Maths – Part 1

Student can practice using the following statements in their group sessions:

I agree because ...  
I like the proposal as ...  
I disagree because  
I do not believe ...  
I like your method because...

## Material and methods

Over the last 40 years, a significant amount of effort has been focused on the importance of problem-based learning (PBL) in collaborative [1, 2] and problem-solving approach [3] as a medium to learning culture in mathematics [4] and a support for collaborative knowledge construction [5]. PBL is a facilitator to enable students to transcend and improve their knowledge and capabilities.



## Talking the Language of Maths – Part 2

Students can further develop their group sessions by using the following methodology:

Read the problem twice and make notes to fully understand the problem at hand.

Each student is provided 2 minutes to explain the problem to their colleagues (this assists students to understand the problem better). Students then characterize the problem and suggest a solution method.

Students take 5-7 minutes to solve the problem individually and if they think they have found a reasonable answer, they share the solution as well as the method. If several fellow students have the same answer, students share their method and let the others who have not solved the problem a few more minutes to answer.

One of the students can solve the problem on the board in the group room and can discuss their views and ideas with their fellow students. Students explain the reason for their choice of method and summarize their understanding.

Students should remember to:

Talk using simple terms, be precise and be short – so the approach remains focused.

Listen to others and their views – in order to fully engage with the process.

Lecturers should remember to:

Makes good questions / problems which lead students to understand definitions and theories with a stronger emphasis.

Listens to the students and try to give them guidance on how they can be:

- more simple and effective;
- identify problems easier; and
- avoid misunderstandings and wrong methods.

Ensure students understand the correct method to apply for certain problems.

## Results

The study provides a good basis for improving students' learning process. This has had positive effects on students' motivation and desire to attend teaching activities and learn effectively.

The talk method contributes to learning by:

Talking simple, short, precise

Read more effective

Listen to others and respect their views

Talking mathematics supports:

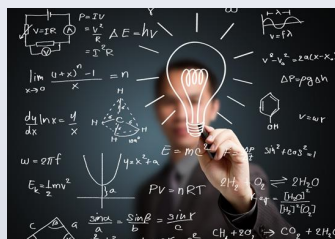
understanding and misunderstanding.

robust learning by boosting memory.

deeper reasoning.

language development.

the development of social skills.



## By talking mathematics

Students learn how to formulate their thoughts in a short and precise way.

Students learn to use diagrams and simple figures to understand and present ideas.

Students better absorb mathematics as their own knowledge and find ways to explain simply complicated problems

Students' comfort level in small groups increases as well as their knowledge base. It gives them greater confidence in their capacity to learn and make sense of mathematics.

The talking method can provide a forum for broader interpretations and an opportunity for students to clarify their understanding.

Students learn to listen to and respect each other's solutions and evaluate different viewpoints.

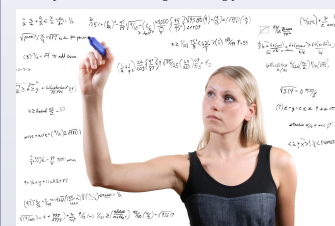
The failure rate for engineering students in first course mathematics is usually around 20-30%.

This method has also been implemented in teaching students from various engineering disciplines:

The mathematics talk application has contributed to reducing the failure rate to 5-10%. Students had a greater enjoyment in attending classes and what could have been long and boring classes turned to be exciting and explorative.

## Conclusions

The result of this case study over seven years indicates that verbal learning process can assist engineering students' knowledge to step from informative knowledge to integrative knowledge. The choice of questions and problems are essential. PBL and TBPS approaches are vital and effective methods to improve students' learning abilities and develop their educational identity and professional skills [6]. The students' ability to gain and retain mathematics knowledge is a key factor for success in their studies. The study confirms the importance of student to student interaction for a deeper and more meaningful learning process.



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